

**QJazz**

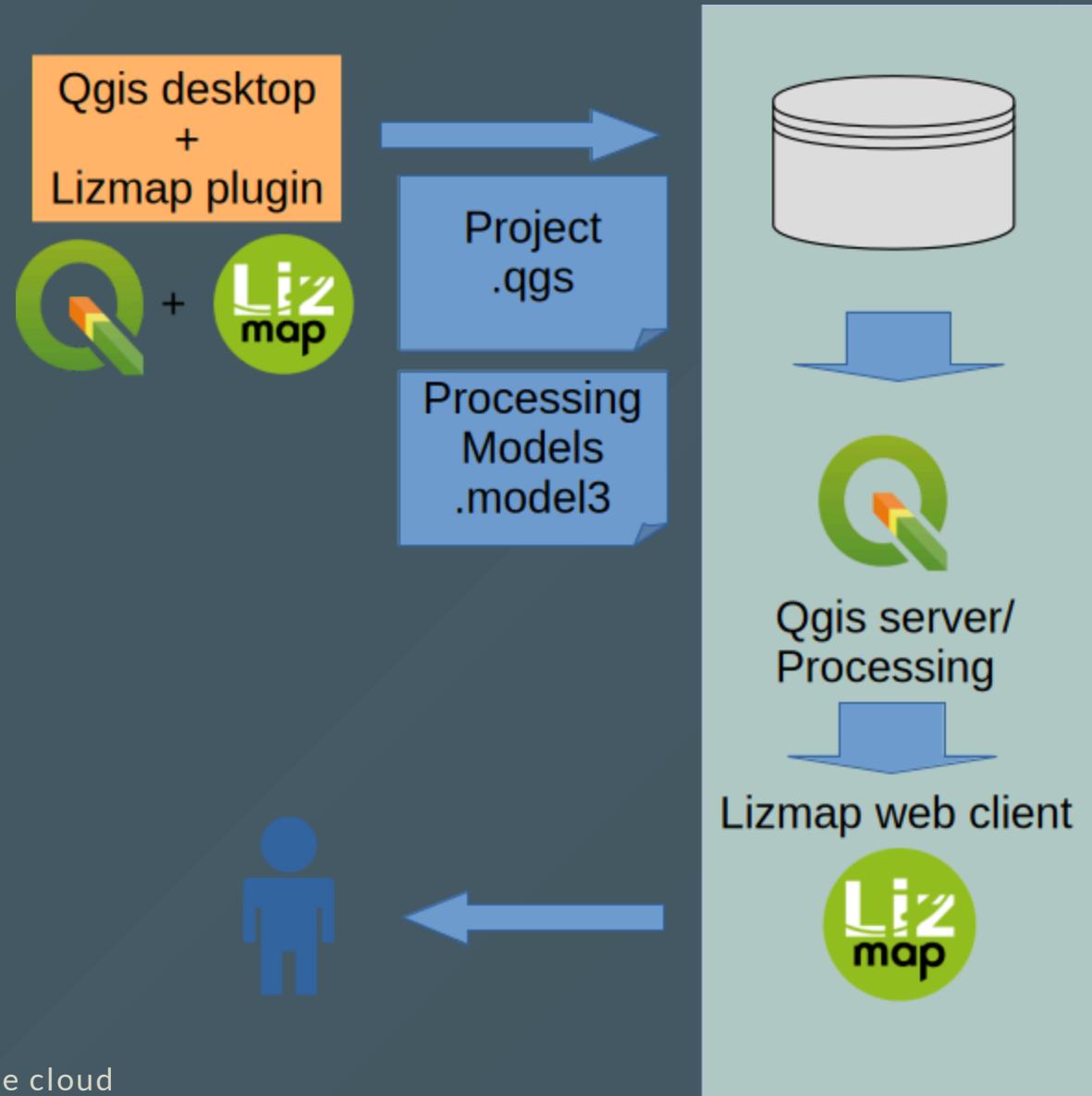


## **QGIS server solution for the cloud**

**Return of experience from GIS hosting services with Lizmap and QGIS server**

<https://github.com/3liz/qjazz>

# GIS hosting services with Lizmap and QGIS server



# GIS hosting services with Lizmap and QGIS server

## Numbers:

- 580 QGIS server instances
- 1500 projects
- 47 physical servers
- 407 PostgreSQL databases
- 377 lizmap instances



# Problems to solve:

- Scalability
- Distributed architecture
- Monitoring
- Zeroconf (ideally !)
- Security

## Dealing with issues

- Projects with many layers: up to 200 layers per project
- Loading times of several minutes
- Memory issues
- High latency requests (mainly remote services)
- Stuck server instances

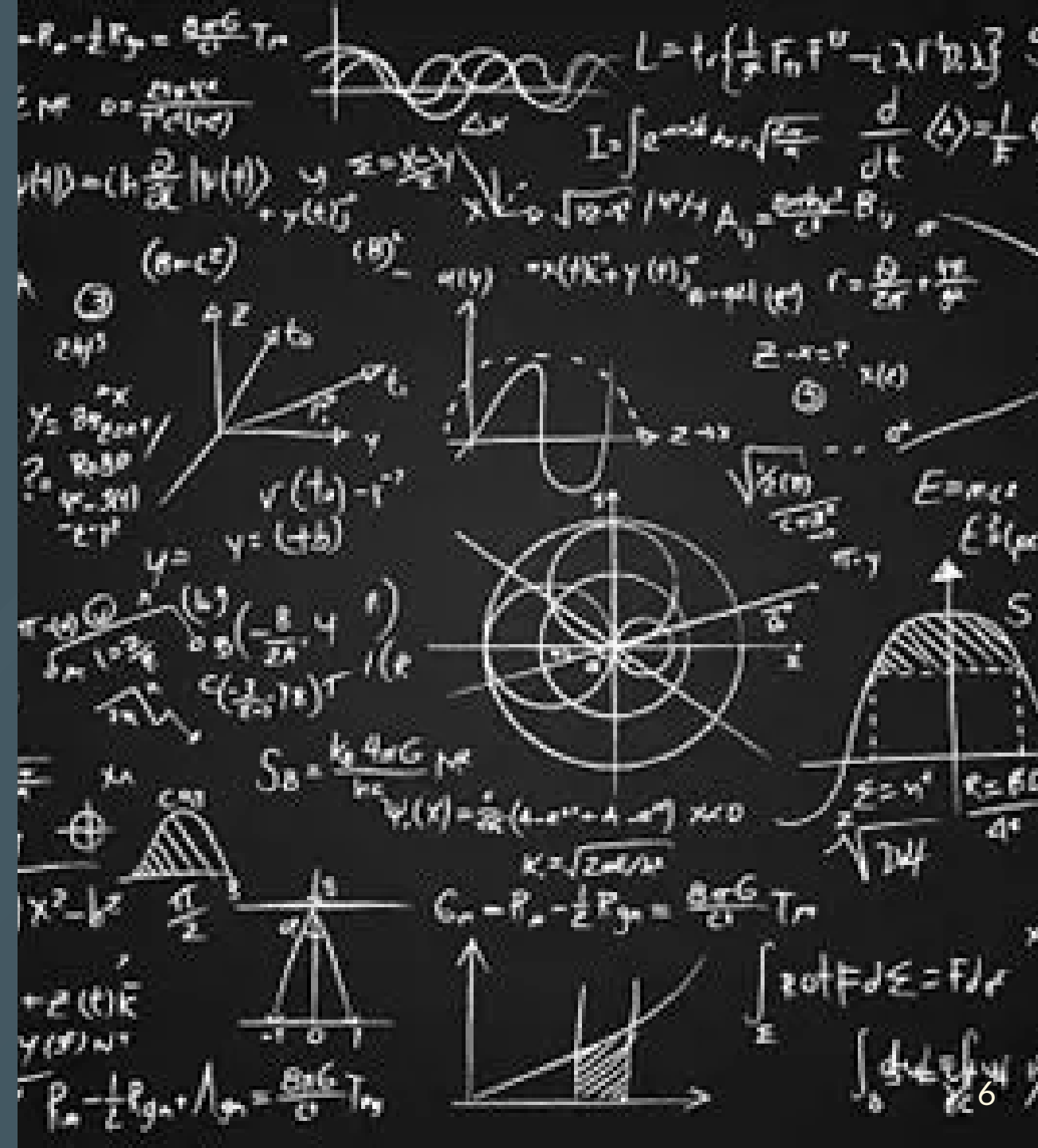
# When things go wrong



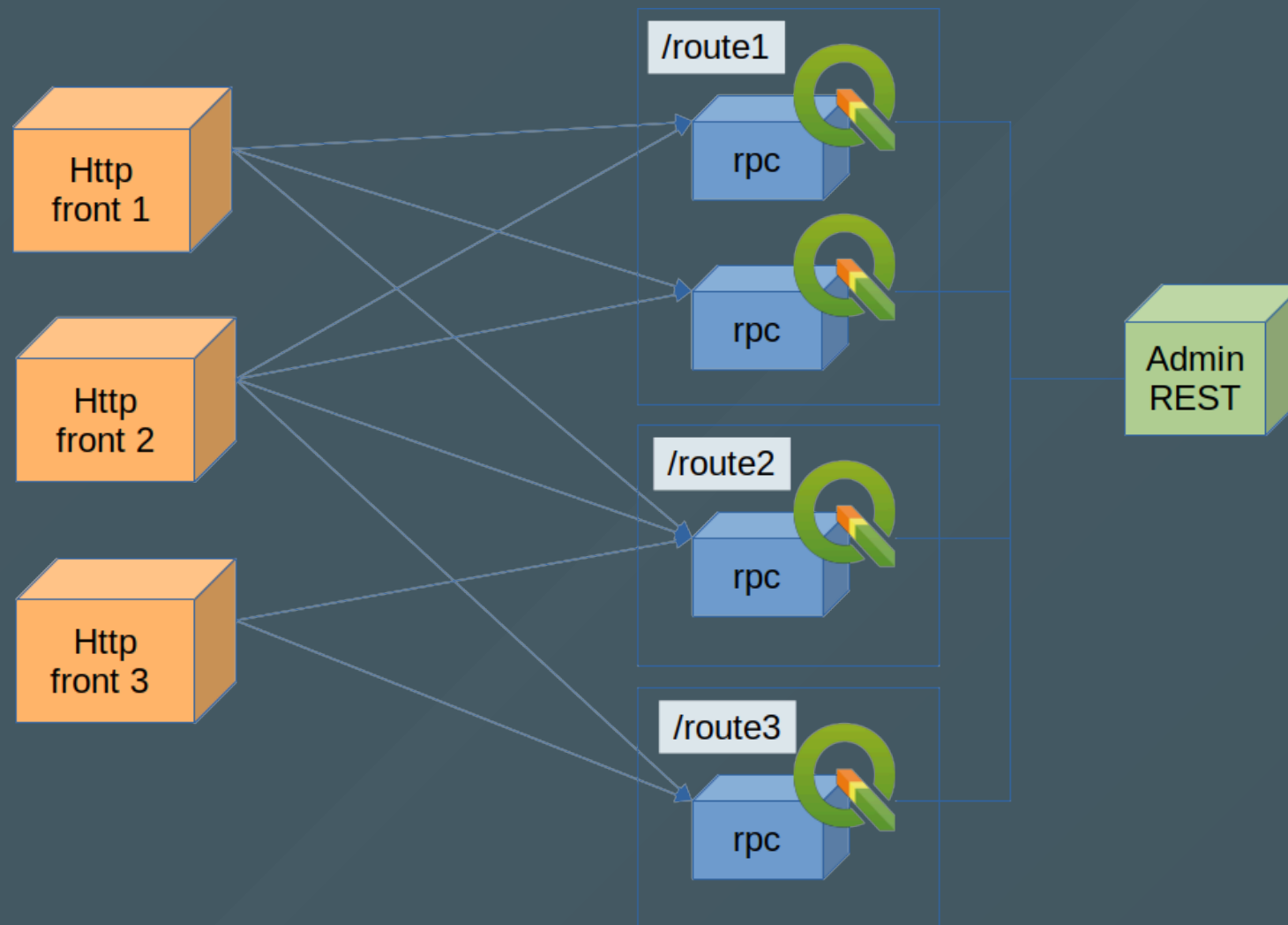
## Microservices with gRPC protocol:

<https://github.com/3liz/qjazz>

- Built-in Load balancing
- Built-in healthcheck support
- Dedicated administration service
- Bi-directional streaming support
- Independent from HTTP front-end



# QJazz overview



# QJazz services

- OGC OWS services (WMS, WFS, ...) - OGIS server native services
- STAC catalogs view of Projects and layers
- OGC API Maps - <https://ogcapi.ogc.org/maps/>
- OGC API Processes - QGIS Processing as a service
- Support for S3 backend (Projects and data)

# QGIS Projects management in QJazz

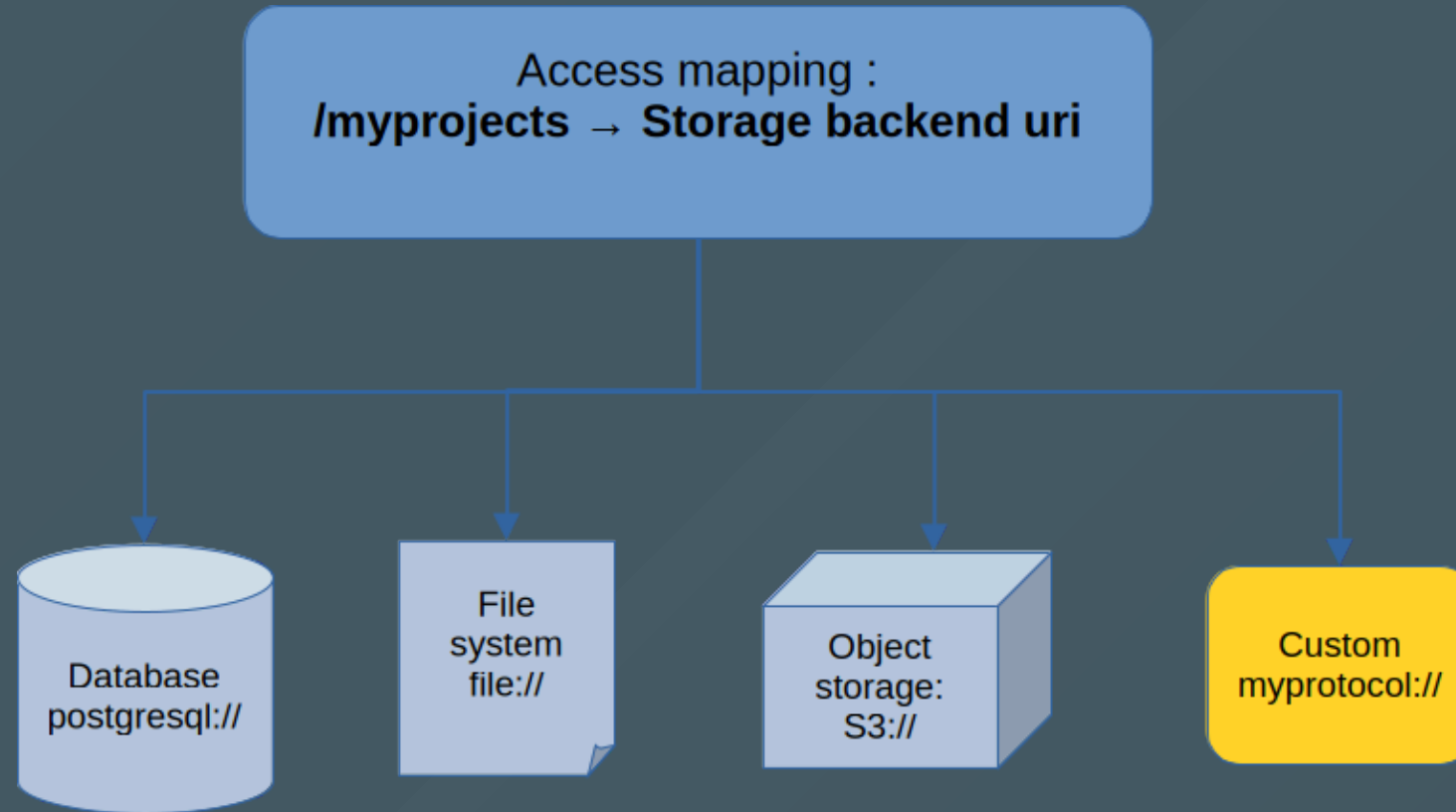
## From a 'project as resource' perspective

- Consider Project as an application
- Corollary: QGIS server is an application server
- Control what is published (from a customer perspective)
- Keep some level of flexibility (dynamic caching)

## From a deployment perspective

- Projects partitioning (routes)
- Handle access controls to backend Apis
- Easy to scale

# Qjazz project's storage



# Performances considerations

## Horizontal scalability governance

- What is the expected request rate ?
- How request distribute on projects ?
- How many different projects I have to handle ?

## Impact on request processing

- How many layers in my project ?
- Data volumetry ?
- Accesses to database backends/external services ?

# There is no single strategy to rule all your projects



# Conclusion

- Revisiting QGIS server as an application server
- Cloud friendly integration (QGIS servers as micro-services)
- Address the problem of serving multiple use cases with many different kind of QGIS projects.

**Thank you**



**Questions ?**